

Inequality and Social Contests

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1 Introduction

The interactions between inequality and status motivations remain contested. While the rise in consumer attention to status goods has been widely critiqued (Veblen, 1899; Frank, 1985), the link between relative status and subjective well-being (Clark, Frijters & Shields, 2008) makes it difficult to dismiss status motivations as socially wasteful. Status-related consumption is not thus merely a reflection of well-established hierarchies - but also a means to build and maintain social hierarchies.

The endogenisation of status in economic models is limited by competing interpretations of hierarchy. Some perspectives explain social hierarchy through unequal access to material resources and structural advantages (Kendall, 2002), whereas others emphasise upon reproduction of inequality through social networks, cultural knowledge, and norms (Weber, 1922; Bourdieu, 1986). The current paper banks upon Weber's (1922) framework of status groups and uses observations from Goode's (1978) commentary on consumption as a mechanism of social mobility to situate status competitions within a closed economy of firms and labour. The economic interactions in the model treat social networks as an essential mechanism to provide skilled labour in the economy.

More specifically, the participants in the closed economy invest in market-relevant adaptability – such as skills, behavioural conformity, and professional signalling – as well as a long-term social categorisation derived from language, nationality, religion, or political identity that also drives success – but evolves far more slowly as it shapes trust, loyalty and the perceived legitimacy of participants within the economy (Tilly, 1998; Bourdieu, 1986). The framework is then used to investigate the extent to which inequality may encourage positional competition and yet present indirect costs through lack of economic growth. The two questions we're concerned with are i) how adaptability related expenditures (or social capital investments) respond to rise in inequality and ii) how inequality imposes limits of growth through constraints of social hierarchy.

2 Model

“People don't live in markets, they live in societies, and the question is whether our society can tolerate the even greater extremes that full internationalization would bring.”

— James Fallows, 1990 *New York Review of Books*, “Wake Up, America!”

At the core of the model is a dichotomy between two forms of social positioning: adaptability and long-term categorisation. Adaptability refers to short-term, market-driven traits such as skills, linguistic abilities, personality fit, and professional behaviour. Such traits are more often shaped by transactional interactions and can be improved through social capital investments influenced by economic incentives.

In contrast to such skills, a certain long-term categorisation mechanism reflects slow-moving social identification including race, religion, political affiliation or nationality. These are shaped through non-transactional, contractual, and trust-based processes of social cohesion and assimilation, and evolve far more slowly than what falls in the realm of adaptability.

This dichotomy reflects a key dynamic in social life: individuals can readily adapt their behaviour to fit social and professional environments (e.g. the “chameleon effect”), but they also must remain constrained by social identities that shape how they interact in a social environment. Even though firms and institutions do not directly allocate resources for these latter mechanisms – relying more often on adaptability to generate revenues – they do operate within broader social environments where long-term categorisation influences trust, loyalty, and stability. These costs are only indirectly borne by firms and institutions.

The model for economy integrates the dichotomy using a framework of labour-queue theory (Spence, 1973). More particularly, the firms in the model must rank workers based on the perceived earning potentials of workers. While the higher rank of an employee may depend on both adaptability and categorisation, the firm’s revenue based planning – which deals more often with short-term pressures – can only consider workers’ adaptability investments. An asymmetry thus follows because of long-term categorisation being of more immediate relevance to workers – the worker invest in both dimensions of social capital: adaptability to enhance short-term productivity and categorisation to secure long-term social positioning whereas firms or institutions only account for adaptability.

The model is operationalised as a three-player game involving a firm and two workers competing in a labour queue (see Section 2.1). The three-game formulation is only a stylised view of a more general N -player economy where workers compete for positions with both social capital investments and social lottery. This microeconomic formulation is detailed in the Appendix.

2.1 Labour and Firm

Drawing on a Schumpeterian closed economy, we assume that all income circulates between workers and firm. The economic growth arises from continual introductions of new or higher-quality goods. Workers supply labour and receive wages – which in turn become the domestic demand. Firms use the revenue generated from this demand – either directly through consumer investment in skills or profit – to fund innovation. Successful innovation produces new product and improves existing services – generating the next round of income. This circular flow of wages, consumption, and innovation creates an endogenous engine of growth: consumption completes one innovation cycle by returning income to firms, while investment initiates the next by financing the search for new products. In long-run equilibrium that follows, the firms choose an optimal allocation of resources that sustain a steady stream of innovation and revenues.

In the three player stylised model, the high-performing and low-performing workers choose an optimal combination of skill investment and the investment in the lottery of social mobility, whereas firms set training costs and wage differentials to maximise asset value net of depreciation as well as asset-maintenance costs. While labour sorting is under firm’s control, the social costs – such as job search costs for workers, or influence of social networks – act as constraints for the firm. The asymmetry between workers and firm is reflected in the fact that while firms capture the returns to innovation and skill formation, the costs of building durable social ties — rooted in long-term categorisation processes — are borne by workers and do not feed back into firm’s revenues.

2.2 Model Dynamics

The choice for workers in the economy is represented as a reference-dependent S-shaped utility function. As detailed in the Appendix, this is implied by an N -player formulation of competitions where

workers use both social lottery and upskilling in a fixed socio-economic ladder. In the stylised view – consisting of two workers and a firm – the workers choose a certain mixing criterion λ to distribute income between social-lottery benefit and the upskilling costs or the social capital investments. The lottery benefit is an exogenous benefit from social connections that is not managed by the firm. We further assume that the training costs charged to consumer are part of exogenous maintenance costs of assets so that all what firm can do in order to maximise its profit in the economy is to adjust the pay-distribution (μ) among the two-workers.

More formally, the firm only controls:

1. μ – the firm chooses $0 < \mu < \frac{1}{2}$ so that the higher-rank (top-performer) worker is given $1 - \mu$ as wage share while the lower-rank (bottom-of-the-queue performer) worker is given μ as wage share. Since μ simultaneously determines the wage split and the degree of income inequality in the closed economy, it serves as both the firm's instrument and the inequality parameter. Workers take μ as given when choosing their upskilling shares λ_L and λ_H .

The differences in choice of the two workers arise solely due to the reference-dependent S-shaped utility curve. In other words, the two workers make different choices for their share of wage spent on social capital investments. The two different selections of mixing variables are:

1. λ_L – portion of wage spent on adaptability by lower-ranked worker
2. λ_H – portion of wage spent on adaptability by higher-ranked worker

2.2.1 Choice for Firm

The growth of firm's assets is subject to asset-maintenance costs $C(A)$ and a constant depreciation δ . The depreciation of assets means that firm are forced to rely on investments from the worker while the asset-maintenance costs pose limits on a steady growth in assets through income addition alone¹. In the long-term equilibrium, a firm chooses μ so that the income stream balances the decline in assets and asset-maintenance costs.

Assuming a constant depreciation δ on accumulated A_t , the firm distributes a fixed proportion π of A_t as wage - while the rest of $A_t(1 - \pi)$ contributes to maintaining or incrementing existing assets. Out of $R_t \equiv \pi A_t$ that needs to be distributed as wages, the firm provides aggregate wages μR_t to lower-ranked workers (mass W) and wages $(1 - \mu)R_t$ to the higher-ranked worker (mass 1). The per-capita wages are then $m_L = \mu R_t / W$ for each lower-rank worker and $m_H = (1 - \mu)R_t$ for the higher-rank worker.

For the low-rank worker's choice of $\lambda_{L,t}$, the per-capita amount $m_L \lambda_{L,t}$ is spent on adaptability, so the aggregate upskilling contribution from all W lower-rank workers is $W m_L \lambda_{L,t} = \mu R_t \lambda_{L,t}$. For the higher-rank worker's choice of $\lambda_{H,t}$, $(1 - \mu)R_t \lambda_{H,t}$ is spent on adaptability. The total contribution $\psi_t = R_t(W \lambda_{L,t} \mu + (1 - \mu) \lambda_{H,t})$ thus flows back to the firm as upskilling investments (workers' expenditure).

Other than the depreciation δA_t , the assets at any given time incur two costs

1. the operational costs with the distribution of $R_t = \pi A_t$ to workers as well as the collection costs for workers' contribution ψ_t – represented as a (power) cost-function $K(\Delta\mu)$
2. the maintenance cost for assets $(1 - \pi)A_t$ – represented as a cost-function $C(A_t)$

The firm thus chooses μ in order to maximise A_{t+1} for a given A_t at every cycle. The dynamics is summarised as follows

¹ All efficiency loss in collection of investments from workers are considered as part of asset-maintaining costs.

$$A_{t+1} = (1 - \pi - \delta)A_t + \pi A_t(W\lambda_{L,t}\mu + (1 - \mu)\lambda_{H,t}) - K(\Delta\mu) - C(A_t) \quad (1)$$

For a fixed $\lambda_{L,t}$ and $\lambda_{H,t}$, a low W causes the firm to benefit from lowering μ since that implies a high collection $(1 - \mu)$ despite lower operational costs (less distribution). In other words, the firm benefits from lower pay so long as $\lambda_{H,t}$ is high and $(1 - \mu)\lambda_{H,t}$ is significant due to low W . This changes only when W and λ_L result in a higher contribution.

The firm arrives at μ by maximising the following *effective upskilling fraction*

$$\phi(\mu) = W\mu\lambda_L^*(\mu) + (1 - \mu)\lambda_H^*(\mu),$$

where $\lambda_L^*(\mu)$ and $\lambda_H^*(\mu)$ are the workers' equilibrium upskilling shares given policy μ , and the weight W reflects the mass of lower-rank workers. The next section explores how μ specifically relates with the selection of $\lambda_{L,t}$ and $\lambda_{H,t}$.

2.2.2 Choice for Workers

We consider a two-level Nash-equilibrium where the mixing parameters λ_L^* and λ_H^* are chosen under a linear utility (underpinning the macro dynamics implied by the N -player stochastic formulation). While the linear utility $u(y) = y$ implies pure risk neutrality, it preserves the essential behavioral asymmetry through quadratic risk premia (see Appendix for details). More particularly, the low-rank player receives a premium for lottery allocation (mimicking risk-seeking / convexity effect) while the high-rank player receives a negative premium (mimicking risk-aversion / concavity effect).

As described in Section 2.2.1, the firm's wage-split μ determines per-capita incomes directly: $m_L = \mu R/W$ for the lower-rank worker and $m_H = (1 - \mu)R$ for the higher-rank worker ($\mu \in (0, 0.5)$). Workers take μ – and hence their incomes m_L and m_H – as given when choosing their upskilling shares. Thereon, the lower-rank and higher-ranked workers choose mixing parameters λ_L and λ_H respectively – as their share of current income allocated to upskilling. The rest of their income is allocated to a shared lottery pool ².

With the linear utility and a linear-approximation for risk-premium, the net income to be optimised by the low-rank worker is

$$y_L = m_L + \eta\lambda_L m_L - C_L(\lambda_L) - (1 - \lambda_L)m_L + \kappa W(1 - \lambda_L)m_L + \rho_L$$

Similarly, the net income optimised by the higher-rank worker is

$$y_H = m_H + \eta\lambda_H m_H - C_H(\lambda_H) - (1 - \lambda_H)m_H + \kappa(1 - \lambda_H)m_H + \rho_H$$

The income components above are defined as follows:

²We discard the approach to use a common lottery benefit to both the workers – as it does not represent the random job-swap dynamics of the N -player formulation. More specifically, the lottery in the N -player formulation is essentially a zero-sum reshuffle of existing Pareto-distributed positions triggered by uniform displacements – wherein a displaced worker pays a cost to draw a random other position, so the gains come directly at others' expense, creating a tight interdependence and rank-dependent externalities that are better represented in the shared-pool abstraction (equivalent to a Tullock lottery with $r = 1$). A fixed common lottery benefit to both workers, on the other hand, introduces exogenous positive-sum value that does not correspond to the displacement of another's position that is a necessary feature of the job-swap feature of the N -player formulation.

$\kappa \approx 1$	mean return per unit contributed to the pool; fairness parameter
W	population-weight assigned to the lower-ranked worker
$\eta \in (0, 1)$	efficiency of upskilling investment
$C_L(\lambda_L) \approx b\eta\lambda_L m_L$	linearized upskilling cost for lower-ranked worker ($b > 1$)
$C_H(\lambda_H) \approx b\eta\lambda_H m_H$	linearized upskilling cost for higher-ranked worker ($b > 1$)
$\rho_L(\lambda_L) = p_L(1 - \lambda_L)^2$	quadratic risk-premium approximation – ($p_L < 0$) proxies mild risk-aversion for the low-rank worker (deterministic penalty for lottery exposure, since $\rho_L < 0$ when $p_L < 0$; weaker than the high-rank penalty since $p_H < p_L < 0$)
$\rho_H(\lambda_H) = p_H(1 - \lambda_H)^2$	quadratic risk-premium approximation – ($p_H < 0$) proxies concavity / risk-aversion (deterministic penalty for lottery exposure, since $\rho_H < 0$ when $p_H < 0$)

It is worth highlighting that the return from lottery investment are based on the aggregate lottery pool :

$$\Pi = W(1 - \lambda_L)m_L + (1 - \lambda_H)m_H$$

Given the efficiency of lottery κ , the lottery return to lower-ranked worker is therefore

$$\kappa \cdot \frac{W(1 - \lambda_L)m_L}{\Pi} \cdot \Pi = \kappa W(1 - \lambda_L)m_L.$$

Similarly, the lottery return to the higher-ranked worker is

$$\kappa(1 - \lambda_H)m_H.$$

Low-rank player's choice

With $p_L < 0$, the second-order condition $\frac{\partial^2 y_L}{\partial \lambda_L^2} = 2p_L < 0$ is satisfied, so the first-order condition identifies a maximum. The first-order condition is

$$\frac{\partial y_L}{\partial \lambda_L} \approx (\eta - b\eta - \kappa W + 1)m_L - 2p_L(1 - \lambda_L) = 0$$

or, rearranging (and noting $\eta - b\eta - \kappa W + 1 = -(\kappa W - \eta + b\eta - 1)$),

$$2p_L(1 - \lambda_L) = -(\kappa W - \eta + b\eta - 1)m_L$$

The low-rank worker thus selects the following interior optimum:

$$\lambda_L^* = 1 - \frac{m_L(\kappa W - \eta + b\eta - 1)}{2p_L}$$

The asymmetry between workers is preserved by the ordering $p_H < p_L < 0$ (i.e. $|p_H| > |p_L|$): the high-rank worker is more risk-averse in lottery exposure than the low-rank worker. This ordering, combined with the larger income $m_H > m_L$ and the absence of the W amplifier in the high-rank FOC, produces $\lambda_H^* > \lambda_L^*$ at equilibrium — the high-rank worker upskills more intensively. We maintain this as a standing assumption:

Assumption A2 (Risk-aversion ordering): $p_H < p_L < 0$.

Since $\lambda_L^* \in (0, 1)$, we bound the solution as:

$$\lambda_L^* = \max\left(0, \min\left(1, 1 - \frac{m_L(\kappa W - \eta + b\eta - 1)}{2p_L}\right)\right)$$

High-rank player's choice

For the high-rank worker, the first-order conditions imply that

$$\frac{\partial y_H}{\partial \lambda_H} \approx (\eta - b\eta - \kappa + 1)m_H - 2p_H(1 - \lambda_H) = 0$$

Therefore, the higher-rank worker selects the following interior optimum:

$$\lambda_H^* = 1 - \frac{m_H(\eta - b\eta - \kappa + 1)}{2p_H}$$

Since $p_H < 0$, the second-order condition $\frac{\partial^2 y_H}{\partial \lambda_H^2} = 2p_H < 0$ confirms this is a maximum. The fraction is positive whenever $(\eta - b\eta - \kappa + 1) < 0$ (upskilling costs and lottery returns sufficiently large), ensuring $\lambda_H^* < 1$. Since $|p_H| > |p_L|$ under Assumption A2, the high-rank worker's risk-aversion dampens lottery allocation more strongly, yielding $\lambda_H^* > \lambda_L^*$ in equilibrium.

$$\lambda_H^* = \max\left(0, \min\left(1, 1 - \frac{m_H(\eta - b\eta - \kappa + 1)}{2p_H}\right)\right)$$

3 Results

This model creates a two-level Nash equilibrium where workers optimise λ_L, λ_H given μ and the firm optimises μ given λ_L^*, λ_H^* . The effective fraction $\phi(\mu)$ is the W -weighted share of wages that becomes productive investment. We maintain the following parameter restriction throughout:

Assumption A1 (Scale condition): $R\kappa > p_L + p_H$.

Since both $p_L < 0$ and $p_H < 0$ under Assumption A2, we have $p_L + p_H < 0$, so Assumption A1 ($R\kappa > p_L + p_H$) is satisfied whenever $R\kappa > 0$, which holds trivially. Its role is thus to ensure the W comparative static is well-signed (as shown below) rather than to impose a binding restriction on parameters.

3.1 Equilibrium value of μ

Since all the terms except $\phi(\mu)$ in Equation 1 are independent of μ , maximising $\phi(\mu)$ (or its steady-state value) directly maximises A_{t+1} and long-run growth³. The fraction is effectively the W -weighted share of total wage spending that ends up as genuine productive investment (upskilling) for the firm (rather than being diverted into the social lottery channel).

The steady-state equilibrium is a fixed point $(\mu^*, \lambda_L^*, \lambda_H^*, A^*)$ satisfying two conditions simultaneously:

1. Given μ^* , both workers choose interior best-response upskilling shares:

$$\lambda_L^* = 1 - \frac{m_L(\kappa W - \eta + b\eta - 1)}{2p_L}, \quad \lambda_H^* = 1 - \frac{m_H(\kappa - \eta + b\eta - 1)}{2p_H}$$

where $m_L = \mu^* R/W$ and $m_H = (1 - \mu^*)R$ (each bounded to $[0, 1]$). Under Assumption A2 ($p_H < p_L < 0$), we have $\lambda_H^* > \lambda_L^*$ at equilibrium.

2. Given λ_L^* and λ_H^* , the firm chooses μ^* to maximise the effective upskilling fraction

³One could view this growth as the endogenous counterpart to the fixed savings rate in a Solow model

$$\phi(\mu) = W\mu\lambda_L^*(\mu) + (1 - \mu)\lambda_H^*(\mu).$$

To find μ^* explicitly, define

$$\bar{A} \equiv \frac{R\kappa}{2p_L}, \quad C \equiv \frac{R(\eta - b\eta + 1)}{2p_L}, \quad B \equiv \frac{R(\kappa - \eta + b\eta - 1)}{2p_H},$$

so that $\lambda_L^*(\mu) = 1 - (\bar{A} - C/W)\mu$ and $\lambda_H^*(\mu) = 1 - B(1 - \mu)$. Substituting into $\phi(\mu)$ and differentiating, the first-order condition $d\phi/d\mu = 0$ yields the closed-form optimum:

$$\mu^* = \frac{(W - 1) + 2B}{2(W\bar{A} - C + B)}.$$

The second-order condition requires $W\bar{A} - C + B > 0$, which holds under mild parameter conditions. The asset level A^* then satisfies

$$A^* = \frac{\pi\phi(\mu^*)}{\delta + c(A^*)^{\gamma-1}},$$

where $C(A) = cA^\gamma$ ($\gamma > 1$).

3.2 Comparative Statics

Table 3.2 summarises directional effects of key parameters on the equilibrium values (holding others fixed).

- The common factor $(b\eta - \eta - \kappa + 1)$ captures the net marginal benefit of moving one unit of income from upskilling to the social lottery.
 - Higher $b\eta$ (more expensive/costly upskilling) $\rightarrow$ favours lottery (and vice versa).
 - Higher κ (more generous lottery return) $\rightarrow$ favours lottery (and vice versa).
- Division by $2p_L$ and $2p_H$ (both negative) scales the interior solutions:
 - $p_L < 0 \rightarrow$ larger $|p_L|$ increases λ_L^* (stronger risk-aversion reduces lottery allocation)
 - $p_H < 0, |p_H| > |p_L| \rightarrow$ larger $|p_H|$ increases λ_H^* more strongly
- **Effect of W on μ^* :** Differentiating the closed-form $\mu^* = \frac{(W-1)+2B}{2(W\bar{A}-C+B)}$ with respect to W gives

$$\frac{d\mu^*}{dW} \propto 2(\bar{A} - C) + 2B(1 - 2\bar{A}) = \frac{R(\kappa - \eta + b\eta - 1)}{p_L} \left(1 - \frac{R\kappa}{p_L}\right) + \frac{R(\kappa - \eta + b\eta - 1)}{p_H}.$$

Letting $\Gamma \equiv \kappa - \eta + b\eta - 1 > 0$ and simplifying, this reduces to $\frac{R\Gamma}{p_L p_H} (p_L + p_H - R\kappa)$. Since $p_L < 0, p_H < 0$, we have $p_L p_H > 0$ and $\Gamma > 0$. Under Assumption A1 ($R\kappa > p_L + p_H$), the factor $(p_L + p_H - R\kappa) < 0$, so $\frac{d\mu^*}{dW} \propto \frac{R\Gamma}{p_L p_H} (p_L + p_H - R\kappa) > 0$ (negative times negative). Hence $\uparrow W \Rightarrow \uparrow \mu^*$: a larger mass of lower-rank workers leads the firm to tilt the wage distribution toward the bottom.

Parameter	μ^* (firm)	λ_L^* (low-rank)	λ_H^* (high-rank)
$\uparrow W$ (larger mass of poor)	$\uparrow$	$\downarrow$	≈ 1 (corner)
$\uparrow \eta$ (better upskilling)	$\uparrow$ or $\downarrow$ (ambiguous)	weakens lottery pull	$\uparrow$
$\uparrow p_L $ (stronger low-rank risk-aversion)	$\downarrow$	$\uparrow$	little change
$\uparrow p_H $ (stronger high-rank aversion)	$\uparrow$	little change	$\uparrow$
$\uparrow \pi$ (higher wage share)	$\uparrow$	little direct effect	little direct effect
$\uparrow \delta$ (higher depreciation)	little direct effect	$\uparrow$ (indirect)	$\uparrow$ (indirect)

Table 1: Directional comparative statics at equilibrium.

4 Discussion

A few implications follow from the model:

- **Mass effect dominates:** Large W pushes the firm toward high μ^* (allocating a large wage share to lower ranks) to leverage the numerous workers. Under Assumption A1, this effect is unambiguous: even modest improvements in λ_L generate large aggregate upskilling when multiplied by the mass W .
- **Sticky top, churning bottom:** The high-rank worker typically stays at $\lambda_H^* \approx 1$ (near-full upskilling), reinforcing stability at the top. Lower-rank workers allocate substantially to the lottery ($\lambda_L^* \ll \lambda_H^*$ under Assumption A2), producing persistent churning at the bottom.
- **Policy relevance:** The firm has a meaningful lever (μ) to counteract baseline inequality. When W is large, optimal policy often favours the bottom, illustrating how institutional wage allocation can influence long-run inequality and growth.
- **Two-level Nash character:** The equilibrium is a mutual best-response fixed point. Small changes in parameters (e.g., stronger risk-aversion $|p_L|$) can shift the entire system through feedback between workers and firm.

Overall, the model shows that when lower ranks are numerous ($W \gg 1$), rational firm behaviour naturally tilts toward high μ^* under Assumption A1, boosting aggregate growth while preserving the micro-level asymmetry of churning at the bottom and stickiness at the top.

Appendix

The closed N -player economy

The stylised model for workers relies on the dynamics implied by an N -player economy where both social lottery and training costs suggest a separation of workers based on risks in the closed economy. In the microeconomic formulation of the endogenous growth economy, the stratified labour is represented as fixed ladder of jobs occupied by workers at any given point in time. Each position in the ladder is associated with a distinct income level that is eventually set by the firm. Every worker holds exactly one job and the number of jobs does not change over time in the economy. While we assume no long-term unemployment, the workers in the economy face temporary job loss ⁴. Upon the job-loss, the worker faces the choice to re-enter the job-market with costs of up-skilling and play a lottery to swap the current position with someone else in the job-market.

In other words, a displaced worker can return to the previous job by investing in the skills or certifications required to perform it. Since higher-ranked jobs with higher-income require greater skill, the upward move requires substantial retraining costs. The retraining costs are forgone by the worker in accepting lower-ranked jobs. Mobility thus occurs through three mechanisms i) voluntary movement down the ladder ii) costly investment to move upward iii) or a lottery for positions among other workers.

The model captures these possibilities with the three choices. The worker may (i) swap her job with a lower-ranked worker who is willing to acquire the skills necessary to move up (therefore paying the price to move up) (ii) pay the skill cost required to claim a higher-ranked position from someone willing to step down, or (iii) pay a fixed cost that provides mobility with some risk (social lottery). Such options reflect different mechanisms of job mobility - training, competition, and luck.

The set up of the model also entails a variation of risk across incomes. More specifically, the higher-income positions have greater training requirements (factoring in a longer search time and other costs arising out of short-term unemployment) and thus a higher acquisition cost. Since workers in high-skill, high-income positions face greater absolute losses when displaced, the risk of income loss in absolute terms increases with higher income-levels. With all workers having the same preferences, the differences in behaviour across income levels are driven by well-known pattern of risk-aversion for small-gains and higher risk-taking for higher losses. This is captured by the model with an S-shaped Friedman-Savage utility for the worker.

Formally, the set up the model includes the following conditions.

1. The choice for the worker is to either pay cost-of-upskilling or use the social lottery to move up the economic ladder. The lottery is of a fixed cost L so that another worker k out of N workers is randomly selected to swap her job with worker i . Worker k does not pay any additional cost other than the fixed lottery cost L for such a swap.
2. If not playing the lottery, a worker with income m_i may take up a job that pays m_k so long as the worker pays a transition-cost $C(m_i, m_k)$ – a power-cost-function defined as follows

$$C(m_i, m_k) = \begin{cases} (m_k - m_i)^b, & \text{for } m_i > m_k \\ 0, & \text{for } m_i < m_k \end{cases}$$

Above ensures that $\forall k \leq i$, we have $C(m_i, m_k) = 0$ i.e. no cost is paid for moving from a position of high-skill to low (or the same) skill. Higher b corresponds to higher costs (charged by the firm - as discussed in Section 2.2).

⁴The technological changes or internal reorganization that lead to such losses circumscribe the creative destruction in endogenous growth economy.

3. The income-levels are specified with a single-parameter a in the pareto income list $x_0(1-i)^{-1/a}$ for $1 \leq i \leq N$. A higher a corresponds to higher-stakes through income levels.

The more a worker upskills oneself, the harder it becomes for the worker to move up (as the swap pool gets limited). A higher-income employee would also have less likelihood of being replaced (less transferrable-skills due to higher costs). The long-term transition-probability can thus be determined from the costs of moving up the ladder.

It follows from the dynamics that in the long-term equilibrium, higher costs (b) discourage mobility and higher stakes (a) encourage bets. The S-shaped utility means that in the long-term equilibrium, the higher-rank workers would prefer the downgrade while the low-rank workers may prefer the lottery.

Linear Approximation for Risk-Premium

Consider a random net income coming from the lottery part:

$$y = \bar{y} + \sigma_y \varepsilon, \mathbb{E}[\varepsilon] = 0, \text{Var}(\varepsilon) = 1$$

Expanding the utility around the mean $\bar{y}$ using a second-order Taylor series, we have

$$u(y) \approx u(\bar{y}) + u'(\bar{y})(\sigma_y \varepsilon) + \frac{1}{2} u''(\bar{y})(\sigma_y \varepsilon)^2$$

Taking expectation:

$$\begin{aligned} \mathbb{E}[u(y)] &\approx u(\bar{y}) + u'(\bar{y}) \cdot 0 + \frac{1}{2} u''(\bar{y}) \sigma_y^2 \\ &= u(\bar{y}) + \underbrace{\frac{1}{2} u''(\bar{y}) \sigma_y^2}_{\text{risk premium}} \end{aligned}$$

This risk premium term $\frac{1}{2} u''(\bar{y}) \sigma_y^2$ is:

- **positive** when $u'' > 0$ (convex region $\rightarrow$ below r) $\Rightarrow$ player behaves as if the lottery return is **higher** than its mean
- **negative** when $u'' < 0$ (concave region $\rightarrow$ above r) $\Rightarrow$ player behaves as if the lottery return is **lower** than its mean

The linear approximation mimics the direction of the N -player formulations effect by adding an incentive for low-rank workers (effective bonus) while penalising high-risk workers. Since σ_y^2 is the variance of the lottery income component, it scales with the share of income allocated to the lottery, $(1 - \lambda)$, rather than the upskilling share λ . For the higher-ranked worker, σ_y^2 can therefore be proxied as

$$\sigma_y^2 \approx c((1 - \lambda_H)m_H)^2 = p_H(1 - \lambda_H)^2 \quad (p_H < 0)$$

Similarly, we can approximate σ_y^2 for the lower-ranked worker as

$$\sigma_y^2 \approx c((1 - \lambda_L)m_L)^2 = p_L(1 - \lambda_L)^2 \quad (p_L < 0)$$